ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET second paper (drawing practical, 80 marks): redraw an adjustable LM317 power supply, draw the circuit diagram of an infrared beam-break sensor from its breadboard layout, and redraw and build a two-transistor LED flasher and a two-hand safety start with an AND-gate IC in draw.io and Fritzing.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Use appropriate application programs: draw.io (or similar) for circuit diagrams and Fritzing for breadboard layouts. If you do not have a computer, draw neatly by hand.
3. Hand in only ONE printout (or page) per question. Every printout must show the question number and your name or examination number.
4. Save your work regularly.

QUESTION 1

Redraw the circuit diagram of the adjustable power supply below using a software tool such as draw.io.

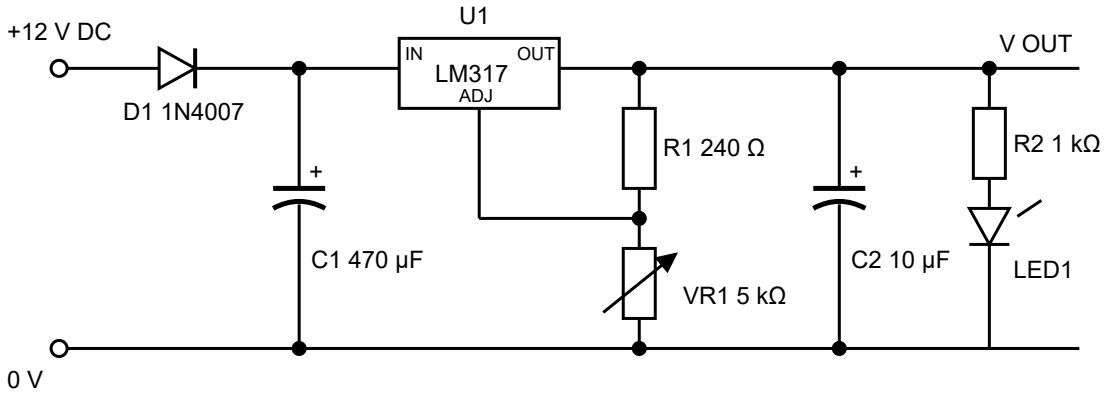


FIGURE 1: Adjustable LM317 power supply

Redraw FIGURE 1 using draw.io (or neatly by hand), with the correct symbols and labels. (10)

[10]

QUESTION 2

A learner built the infrared beam-break sensor described in the table below on a breadboard. LED2 lights up when something blocks the infrared beam. Draw the CIRCUIT DIAGRAM of this layout using a software tool such as draw.io.

Component	Connections on the breadboard
6 V battery	+ to the top (+) rail; − to the bottom (−) rail
150 Ω resistor	from the + rail to row 5
IR LED (LED1)	anode in row 5, cathode in the − rail
10 kΩ resistor	from the + rail to row 12
Phototransistor	collector in row 12, emitter in the − rail (facing the IR LED)
4,7 kΩ resistor	from row 12 to row 18

BC547 transistor	collector row 24, base row 18, emitter in the − rail
330 Ω resistor	from the + rail to row 28
Red LED (LED2)	anode in row 28, cathode in row 24

Draw the circuit diagram of the breadboard layout described above. (10)

[10]

QUESTION 3

3.1 Redraw the circuit diagram of the two-transistor LED flasher below using draw.io. The two LEDs flash on and off one after the other. (15)

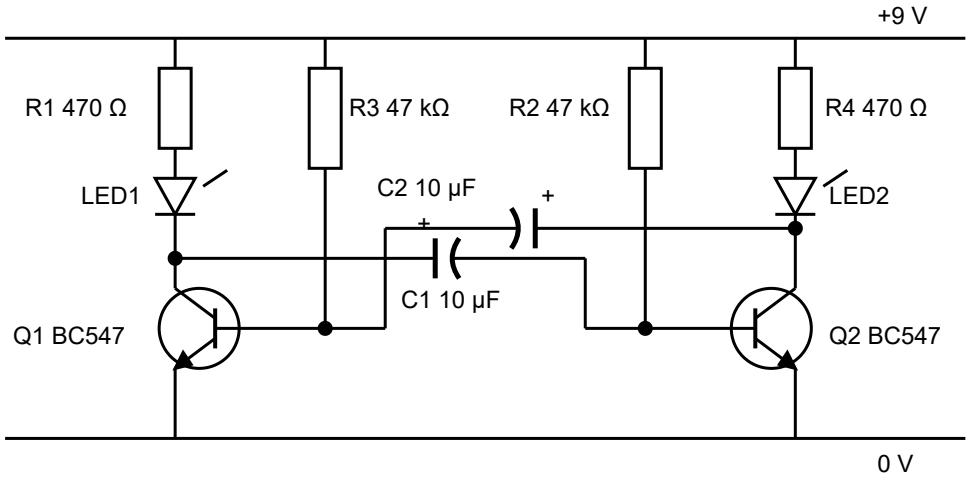


FIGURE 2: Two-transistor LED flasher (lines that cross without a dot are not connected)

3.2 Using Fritzing, create the breadboard layout of the LED flasher in FIGURE 2. (15)

[30]

QUESTION 4

4.1 Redraw the circuit diagram of the two-hand safety start below using draw.io. The machine-ready LED comes on only while BOTH buttons are pressed. (15)

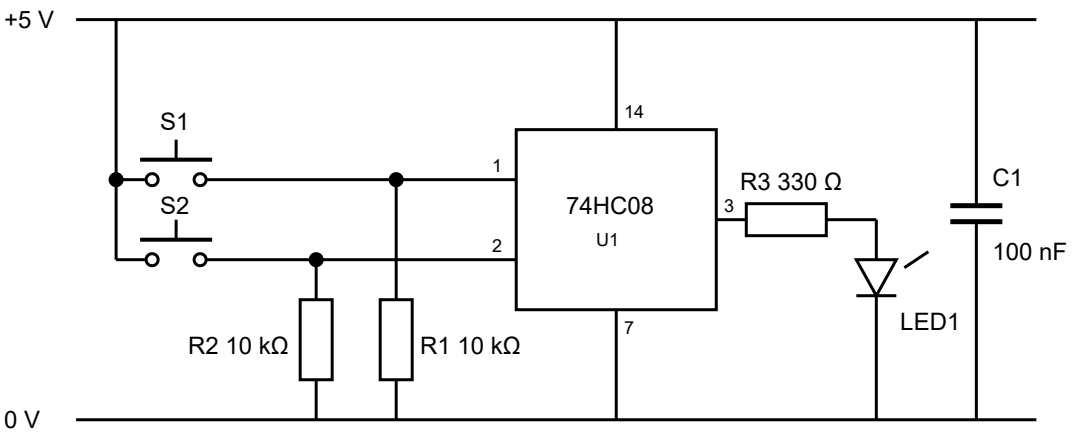


FIGURE 3: Two-hand safety start with an AND gate (lines that cross without a dot are not connected)

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4.2 Using Fritzing, create the breadboard layout of the safety start circuit in FIGURE 3. (15)

[30]

TOTAL: 80

END OF QUESTION PAPER